

Module 3: IAM Policies Assignment

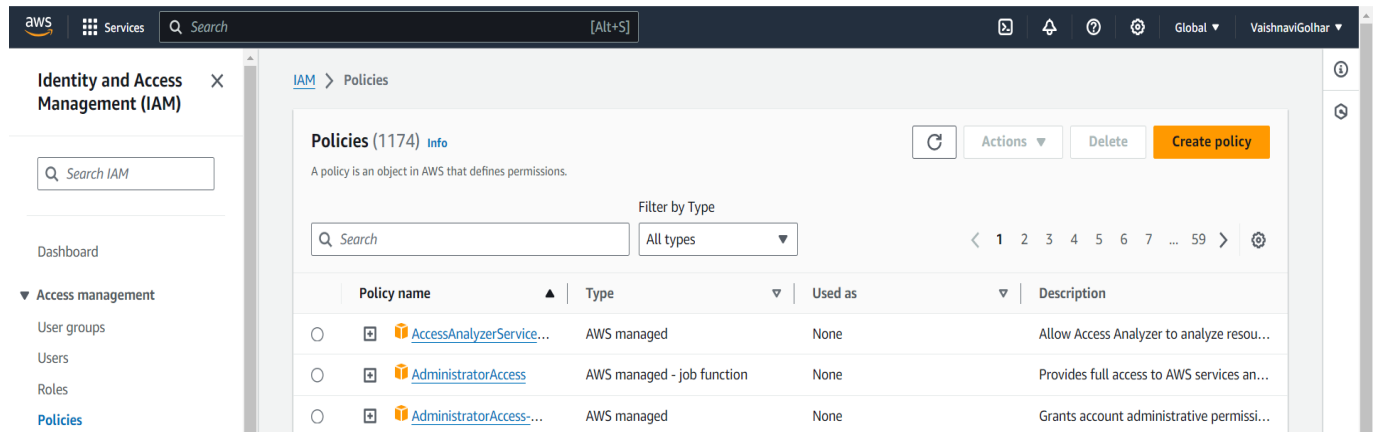
Problem Statement: You work for XYZ Corporation. To maintain the security of the AWS account and the resources you have been asked to implement a solution that can help easily recognize and monitor the different users.

Tasks To Be Performed:

- 1. Create policy number 1 which lets the users to:**
 - a. Access S3 completely**
 - b. Only create EC2 instances**
 - c. Full access to RDS**
- 2. Create a policy number 2 which allows the users to:**
 - a. Access CloudWatch and billing completely**
 - b. Can only list EC2 and S3 resources**
- 3. Attach policy number 1 to the Dev Team from task 1**
- 4. Attach policy number 2 to Ops Team from task 1**

Solution:

1. Go to the IAM Management Console.
2. Choose "Policies" on the left sidebar, click on create policy.



3. Create Policy-1 which lets the users to:

- a. Access S3 completely
- b. Only create EC2 instances
- c. Full access to RDS.

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Services

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IAM > Policies > Create policy

Step 1
Specify permissions

Step 2
Review and create

Specify permissions

Info

Add permissions by selecting services, actions, resources, and conditions. Build permission statements using the JSON editor.

Policy editor

VisualJSONActions

S3

AllowAll actions

Specify what actions can be performed on specific resources in S3.

Actions allowed

Specify actions from the service to be allowed.

Filter Actions

Manual actions | Add actions

☒ All S3 actions (s3:*)

Access level

List (Selected 15/15)

Read (Selected 60/60)

Effect

☒ Allow☐ Deny

Expand all | Collapse all

aws

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Read (Selected 60/60)

Write (Selected 56/56)

Permissions management (Selected 15/15)

Tagging (Selected 12/12)

Required permissions not selected.

To grant permissions for the selected resource actions, you must include additional required actions

- s3:CreateJob requires 1 more action.
- s3:PutReplicationConfiguration requires 1 more action.

Resources

Specify resource ARNs for these actions.

☒ All☐ Specific

The all wildcard "*" may be overly permissive for the selected actions. Allowing specific ARNs for these service resources can improve security.

Request conditions - optional

Actions on resources are allowed or denied only when these conditions are met.

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Services

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[Alt+S]

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EC2

Allow

1 Action

Specify what actions can be performed on specific resources in EC2.

Actions allowed

Specify actions from the service to be allowed.

runl

Effect

Allow

Deny

Write

RunInstances

Info

Required permissions not selected.

To grant permissions for the selected resource actions, you must include additional required actions

ec2:RunInstances requires 3 more actions.

Resources

Specify resource ARNs for these actions.

All

Specific

The all wildcard '*' may be overly permissive for the selected actions. Allowing specific ARNs for these service resources can improve security.

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RDS

Allow

All actions

Specify what actions can be performed on specific resources in RDS.

Actions allowed

Specify actions from the service to be allowed.

Filter Actions

Effect

Allow

Deny

Manual actions | Add actions

All RDS actions (rds:*)

Access level

List (Selected 45/45)

Read (Selected 5/5)

Write (Selected 113/113)

Permissions management (Selected 1/1)

Tagging (Selected 2/2)

Expand all | Collapse all

Services
Search
[Alt+S]

Edit

Permissions defined in this policy [Info](#)

Permissions defined in this policy document specify which actions are allowed or denied. To define permissions for an IAM identity (user, user group, or role), attach a policy to it

Search

Allow (3 of 404 services)
Show remaining 401 services

Service	Access level	Resource	Request condition
EC2	Limited: Write	All resources	None
RDS	Full access	All resources	None
S3	Full access	All resources	None

Add tags - optional [Info](#)

Tags are key-value pairs that you can add to AWS resources to help identify, organize, or search for resources.

No tags associated with the resource.

Add new tag

You can add up to 50 more tags.

Cancel Previous Create policy

Services
Search
[Alt+S]

Identity and Access Management (IAM)

Dashboard
Access management
Users
Roles
Policies
Identity providers
Account settings
Access reports
Access Analyzer
External access
Unused access
Analyzer settings
Credential report
Organization activity
Service control policies (SCPs)

IAM > Policies > Policy-1

Delete

Policy-1 [Info](#)

Policy details

Type	Creation time	Edited time	ARN
Customer managed	February 22, 2024, 21:16 (UTC+05:30)	February 22, 2024, 21:16 (UTC+05:30)	arn:aws:iam::161681788148:policy/Policy-1

Permissions
Entities attached
Tags
Policy versions (1)
Access Advisor

Permissions defined in this policy [Info](#)

Permissions defined in this policy document specify which actions are allowed or denied. To define permissions for an IAM identity (user, user group, or role), attach a policy to it

Copy Edit Summary JSON

```

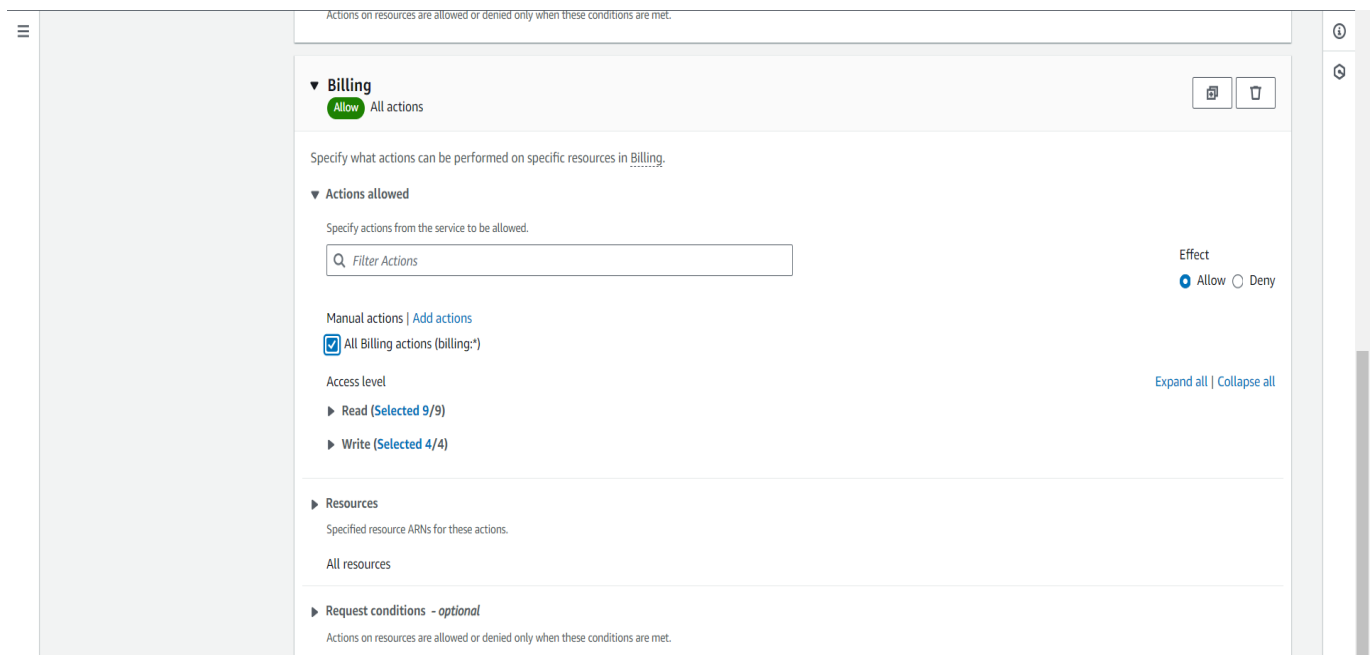
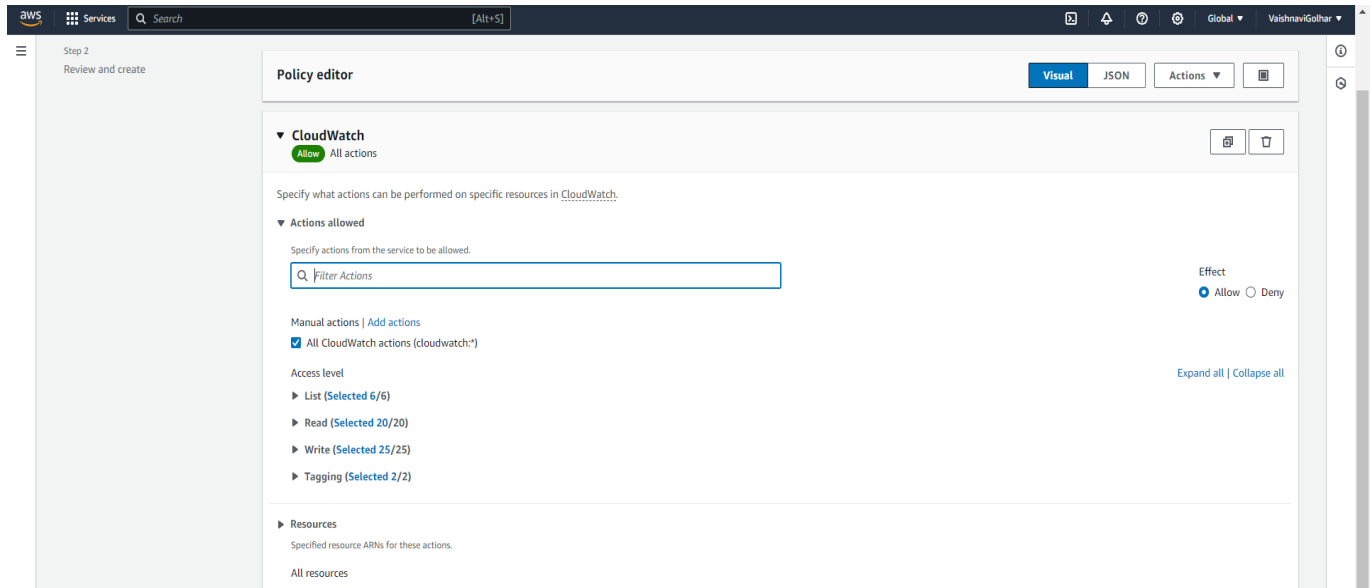
1 {
2   "Version": "2012-10-17",
3   "Statement": [
4     {
5       "Sid": "VisualEditor0",
6       "Effect": "Allow",
7       "Action": [
8         "rds:*",
9         "s3:*",
10        "ec2:RunInstances"
11      ],
12       "Resource": "*"
13     }
14   ]
15 }

```

4. Create a Policy-2 which allows the users to:

a. Access CloudWatch and billing completely

b. Can only list EC2 and S3 resources



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Services

Search

[Alt+S]

Global

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EC2

Allow

172 Actions

Specify what actions can be performed on specific resources in EC2.

▼ Actions allowed

Specify actions from the service to be allowed.

Q

Filter Actions

Manual actions | [Add actions](#)

☐ All EC2 actions (ec2:*)

Access level

► List (Selected 172/172)

► Read (35)

► Write (417)

► Permissions management (5)

► Tagging (2)

Effect

☒ Allow ☐ Deny

[Expand all](#) | [Collapse all](#)

⚠ Required permissions not selected.

To grant permissions for the selected resource actions, you must include additional required actions

- ec2:DescribeLaunchTemplateVersions requires [1 more](#) action.

▼ Resources

Specify resource ARNs for these actions.

☒ All ☐ Specific

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Services

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S3

Allow

15 Actions

Specify what actions can be performed on specific resources in S3.

▼ Actions allowed

Specify actions from the service to be allowed.

Q

Filter Actions

Manual actions | [Add actions](#)

☐ All S3 actions (s3:*)

Access level

► List (Selected 15/15)

► Read (60)

► Write (56)

► Permissions management (15)

► Tagging (12)

Effect

☒ Allow ☐ Deny

[Expand all](#) | [Collapse all](#)

⚠ The all wildcard "*" may be overly permissive for the selected actions. Allowing specific ARNs for these service resources can improve security.

► Request conditions - optional

Actions on resources are allowed or denied only when these conditions are met.

Review and create [Info](#)

Review the permissions, specify details, and tags.

Policy details

Policy name
Enter a meaningful name to identify this policy.
Policy2
Maximum 128 characters. Use alphanumeric and '+-,@-_' characters.

Description - optional
Add a short explanation for this policy.

Maximum 1,000 characters. Use alphanumeric and '+-,@-_' characters.

Permissions defined in this policy [Info](#) [Edit](#)

Permissions defined in this policy document specify which actions are allowed or denied. To define permissions for an IAM identity (user, user group, or role), attach a policy to it.

Allow (4 of 404 services) [Show remaining 400 services](#)

Service	Access level	Resource	Request condition
Billing	Full access	All resources	None
CloudWatch	Full access	All resources	None
EC2	Limited: List	All resources	None
S3	Full: List	All resources	None

Add tags - optional [Info](#)

Tags are key-value pairs that you can add to AWS resources to help identify, organize, or search for resources.

No tags associated with the resource.

[Add new tag](#)
You can add up to 50 more tags.

[Cancel](#) [Previous](#) [Create policy](#)

5. Now 2 Policies has been created.

Identity and Access Management (IAM)

Policies (1176) [Info](#) [Refresh](#) [Actions](#) [Delete](#) [Create policy](#)

A policy is an object in AWS that defines permissions.

Filter by Type: [Customer managed](#) 2 matches

Policy name	Type	Used as	Description
☐ Policy-1	Customer managed	None	-
☐ Policy2	Customer managed	None	-

6. Attach policy number 1 to the Dev Team from task 1

The screenshot shows the AWS IAM console interface. The left sidebar contains the navigation menu with sections like 'Access management' and 'Access reports'. The main content area is titled 'DevTeam' and shows the 'Permissions' tab. Under 'Permissions policies (0)', it states 'You can attach up to 10 managed policies.' and shows a table with no resources to display.

DevTeam Summary:

- User group name: DevTeam
- Creation time: February 22, 2024, 12:45 (UTC+05:30)
- ARN: arn:aws:iam::161681788148:group/DevTeam

Permissions policies (0):

You can attach up to 10 managed policies.

Policy name	Type	Attached entities
No resources to display		

The screenshot shows the AWS IAM console interface after attaching a policy. A green banner at the top indicates 'Policies attached to this user group.' The 'Permissions policies (1)' section now shows one policy attached.

DevTeam Summary:

- User group name: DevTeam
- Creation time: February 22, 2024, 12:45 (UTC+05:30)
- ARN: arn:aws:iam::161681788148:group/DevTeam

Permissions policies (1):

You can attach up to 10 managed policies.

Policy name	Type	Attached entities
Policy-1	Customer managed	1.

7. Attach policy number 2 to Ops Team from task 1

The screenshot displays the AWS IAM console interface. On the left, the 'Identity and Access Management (IAM)' sidebar is visible, with 'User groups' selected under 'Access management'. The main content area shows the 'OpsTeam' user group details. A green banner at the top indicates 'Policies attached to this user group.' Below this, the 'OpsTeam' group is shown with a 'Delete' button. The 'Summary' section provides details: User group name 'OpsTeam', Creation time 'February 22, 2024, 12:46 (UTC+05:30)', and ARN 'arn:aws:iam::161681788148:group/OpsTeam'. The 'Permissions' tab is active, showing 'Permissions policies (1)' with a search bar and a 'Filter by Type' dropdown set to 'All types'. A table lists the attached policy:

☐	Policy name ↗	Type	Attached entities
☐	Policy2	Customer managed	1

Buttons for 'Simulate', 'Remove', and 'Add permissions' are located above the table. The bottom of the console shows a breadcrumb trail: 'IAM > User groups > OpsTeam'.

